



Urban heat stress and human health in Bangkok, Thailand

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ABSTRACT

Heat stress has been recognized as one of the consequences of climate change in urban areas. Its adverse effects on the urban population range from economy, social, environment, and human health. With the increasing urbanization and economic development in cities, heat stress is expected to worsen. This particular study aims to achieve two objectives: (1) to understand the determinants of heat stress, especially the roles of the urban environment in exacerbating the heat stress, and (2) to explore the effects of heat stress to human health using self-reported health assessment. We employed a cross-sectional study using a survey questionnaire from 505 respondents living in the urban area of Bangkok, Thailand. We found that socioeconomic conditions of the individual and urban environment were significant determinants of urban heat stress. Low-income urban populations living in high-density areas with less green open space were more likely to experience heat stress. We also found that heat stress significantly affects human health. Those who reported a higher level of heat stress were more likely to have adverse health and well-being outcomes. The findings suggest that the increased risk of heat stress represents a major problem in the Bangkok, Thailand. It is necessary to address heat stress in adaptation policy and measures at the city levels amid the continued increase of global temperature and climate change.

1. Introduction

Evidence has shown that climate change has significant impacts on human health (Watts et al., 2018). Increasing global surface and ambient temperature are considered as one of the main factors in causing substantial health burden globally (Guo et al., 2017). Many prevalent human diseases are attributable to climate change, from cardiovascular and respiratory diseases to altered transmission of infectious diseases and malnutrition (Patz et al., 2005). The consequences of increasing global temperature and climate change include the raise occurrences of heat stress, heat waves, and higher intensity of urban heat island (UHI) (Basara et al., 2010).

Increasing heat stress has substantial adverse effects on the economy, ecosystem, infrastructure, and human health (Wouters et al., 2017). Its adverse effects on human health range from decreased well-being to the increasing prevalence of heat-related illnesses and mortality (Hatvani-Kovacs et al., 2016). Heat stress can also have a significant influence on daily human activities. Excess heat exposures during daily activities can cause harmful effects to physical health, such as heat exhaustion, heat strokes, and heat cramps (Lemonsu et al., 2015). Heat stress also affects mental health by causing mood changes and increase physiological distress (Salata et al., 2017). Empirical

evidence examining heat stress mainly uses two methods. First is by establishing a numerical model of heat stress as a function of temperature, humidity, radiation, wind speed, and human physiological factors. (for example, see Gao et al., 2018; Chapman et al., 2017; Oleson et al., 2015). This method involves spatial and statistical modeling and climate-model projection to understand how heat stress is affecting cities spatially (Wouters et al., 2017). The second method is by conducting a self-rated health assessment through a survey questionnaire (Schuster et al., 2017). Studies using this method assess self-reported heat stress based on survey questionnaire of the population (for example, see Zander et al., 2018a; Zander et al., 2019a; Harlan et al., 2006; Tawatsupa et al., 2010). This method is considered influential in overcoming the knowledge gaps in multidisciplinary urban heat stress research (Schuster et al., 2017).

Urban areas are particularly at risk of heat stress. Evidence has found to suggest that heat stress increase in cities are two times higher than in rural areas (Wouters et al., 2017). Enhanced urban temperature, humidity, radiation, winds, air pollution, urban heat island, and global warming exacerbate the heat stress effects and deprive the health and well-being of urban residents (Park et al., 2018). Previous research identified age, gender, education, and having chronic diseases were strongly associated with urban heat stress and mortality (Kovacs and

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Hajat, 2008). Other studies reported that environmental factors such as housing conditions, availability of green space, and population density were also significantly correlated with individual heat stress risks in the urban area (Zander et al., 2019b; Harlan et al., 2006).

Negative impacts of heat stress on human health have been studied mainly in European and US cities (Wouters et al., 2017; Lemonsu et al., 2015). Studies in the context of Asian cities with a tropical climate are still scarce. To date, most research in this region has focused on occupational heat stress using population-based assessment (Gao et al., 2018). An examination of the determinants using individual data is necessary to provide a better understanding of the effects of heat stress on health and well-being in the urban areas (Schuster et al., 2017). Furthermore, increasing UHI intensity caused by urbanization and economic development in Asian cities with tropical climate may reveal different patterns and characteristics of the effects of heat stress. For example, a study in the Philippines found that heat stress increases with population density in the urban area (Zander et al., 2018a). The same research also found that air conditioning did not relieve heat stress in high-density urban areas.

The present study is designed to expand the understanding of heat stress in the urban area of Bangkok, Thailand. The study hypothesis is that urban heat stress at the individual level is influenced by urban environment determinants, health behaviors, and socioeconomic conditions. Previous studies in the country have been focused on occupational hazards of heat stress using population data (Tawatsupa et al., 2012; Huang et al., 2018; Thawillarp, 2015). The two objectives of the study are: (1) to understand the determinants of heat stress, especially the roles of the urban environment in exacerbating the heat stress, and (2) to explore the effects of heat stress to human health using self-reported health assessment. This study will add to the limited literature of heat stress in Thailand in particular and Asian cities with a tropical climate.

2. Materials and methods

2.1. Study site

Bangkok is the capital of Thailand and inhabited by more than 8 million people in an area of 1569 sq. km (Arifwidodo and Chandrasiri, 2013). As the capital city and center of industries, manufacture, economy, commerce, and construction of the country, it has been experiencing increasing temperatures and a higher incidence of UHI (Arifwidodo, 2015). In 2015, the maximum temperature difference between the urban and rural areas of Bangkok was 7 °C, which is highest than in the last ten years, and the prevalence of temperature-related mortality and heat-related illnesses such as heat stress has also been increasing (Arifwidodo et al., 2019).

2.2. Survey design

The study is a cross-sectional, self-reported survey questionnaire to understand the determinants of heat stress and its impact on health outcomes in the urban area of Bangkok, Thailand. We selected proxy variables for the study that have been acknowledged in the literature to be associated with heat stress. Fig. 1 summarizes the conceptual framework of the study.

Previous studies have established that assessing heat stress using different domains is the standard method to explain its effect on human health in different population groups (EPSTEIN and MORAN, 2006; Tawatsupa et al., 2010). In this study, urban heat stress was identified as an uncomfortable feeling when doing daily activities as a consequence of heat using four different domains: sleeping, daily travel, work, and exercise. A similar measurement was used by previous studies in Thailand and the Philippines (Tawatsupa et al., 2012; Zander et al., 2018a). Each domain of urban heat stress was assessed by a five-point Likert scale and measured using the proxy questions representing

four domains: 'How often did the hot period of this year interfere with the activities of sleeping, housework, daily travel, work, and exercise?' In the questionnaire, respondents were required to reply to the agreement to the statement, "I have trouble sleeping in the last week because it was too hot," with five possible answers: a little or none, slight, moderate, quite a lot, and almost every day. We made this approach for each domain of heat stress.

We used three proxy variables to explain different aspects of health outcomes: (1) life satisfaction, (2) the energy level for daily activity, and (3) emotional problem. These outcomes were selected because they represented the holistic definition of health by WHO as 'a complete state of physical, mental and well-being and not merely the absence of disease or infirmity' (Card, 2017). The proxy questions for these health outcomes were adopted from the medical outcomes short-form instrument (SF-8), a generic multipurpose short-form health-related quality of life questionnaire (Lang et al., 2018). This form is widely known in public health discipline and has been used in previous studies related to self-reported heat stress and human health (Tawatsupa et al., 2012, 2014; Kjellstrom et al., 2018). Life satisfaction was measured using the question: 'Overall, how would you rate your life satisfaction during the past four weeks?' (Very good, good, fair, poor, or very poor). For the analysis purposes, the last two categories were grouped into 'poor life satisfaction.' Energy level for daily activity proxy question was: 'How often did you have a problem with your energy level due to heat in the past for weeks?' and Emotional: 'How often did you have a problem with your emotion due to heat in the past for weeks?' with responses ranged from 'a little or none, slightly, moderate, quite a lot, and almost every day.' For the analysis, we combined the last two responses with 'having problems.'

Urban environment determinants in this study were conceptualized as the characteristics of the built environment where the respondents lived (Harlan et al., 2006; Schuster et al., 2017). In linking the urban environment to heat stress, we selected proxy variables based on previous studies: (1) air-conditioning at home and work, (2) district population density, (3) proximity to green spaces, and (4) UHI intensity. Literature has suggested that having air-conditioning equipment in work and home can reduce the effect of heat stress (Zander et al., 2018b). Living in a high-density area also considered can increase the exposure to heat stress (Erell et al., 2014). Having green open spaces around the home can reduce vulnerability to heat stress (Ekkel and de Vries, 2017). Previous studies have found to suggest that health behaviors were associated with heat stress at the individual level, which includes having a chronic disease and regular cigarette or alcohol consumption (Kenny et al., 2010). Urban heat island (UHI) has been acknowledged in the literature as one of the major determinants of heat stress in previous studies (Hosking and Campbell-Lendrum, 2012; Dang et al., 2017; Phung et al., 2016). It must be noted that there were some limitations in the selection of variables to link urban environment and heat stress due to data unavailability. For example, we could not include other essential variables of the urban environment, such as road infrastructures, pedestrian conditions, building materials, and air pollution, since the data were not available at the individual level.

2.3. Data collection and sampling

We used a quota sampling design for primary data collection. We divided the city into four areas based on the population density of the city: (1) the downtown area or central business district (CBD), (2) the high-density residential area, (3) the medium and low-density residential area, (4) the sub-urban area. We followed previous studies in Bangkok, which found that urban temperature difference is significantly associated with urban land use and population density (Arifwidodo and Tanaka, 2015; Arifwidodo, 2015). The study then obtained a list of registered households and their addresses from the Bangkok Metropolitan Administration (BMA) and randomly selected 130 respondents for each area (520 respondents in total). A similar

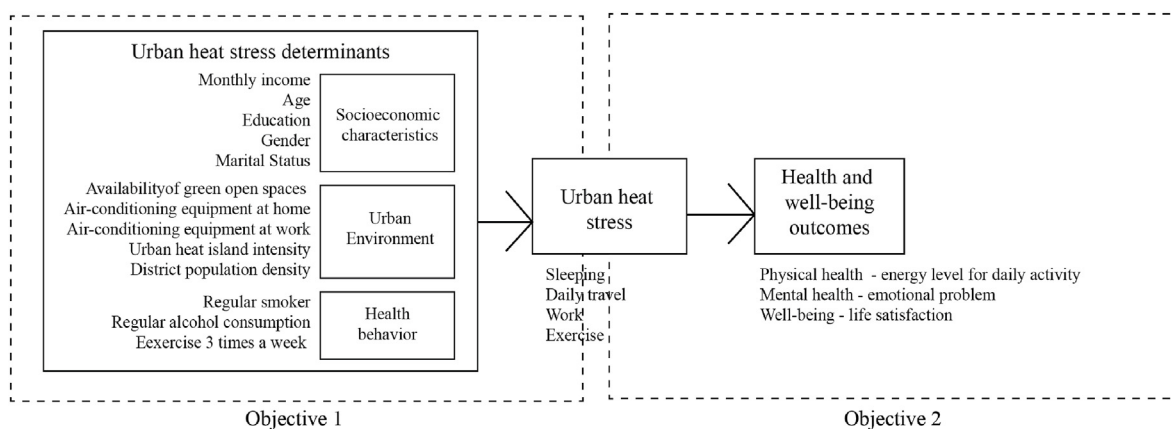


Fig. 1. Conceptual Framework of the study.

method was used by previous studies to obtain a smaller sample in a metropolitan area without creating a larger bias in the data collection (Honold et al., 2012; Schuster et al., 2017). Trained surveyors were assigned to conduct face-to-face interviews from every respondent to ensure the 100% response rate. We surveyed between 10 July – 31 August 2016, during the hot and humid season in Bangkok. After we conducted data cleaning, 505 responses were used for the analysis.

The questionnaire collected other variables such as health behavior and socioeconomic and demographic background of the respondents. We obtained the daily air temperature during the survey period from The Meteorological Department in Bangkok (three weather stations) and Pathumthani Province (one station) to compare the urban-rural condition. These stations were selected because of their proximity to four urban areas of Bangkok that we used for the sampling design

(Fig. 2). We then created a proxy variable called 'UHI intensity', which was defined as the yearly maximum temperature difference between urban area and sub-urban area following the definition used in the previous study (Tan et al., 2010). We also collected the district population density from the BMA. We then geocoded the respondent's home address to incorporate these variables into their responses. For the analysis purpose, we categorized these variables into 'low,' 'medium,' and 'high.' Fig. 2 shows the study area and corresponding information related to the data collection method.

2.4. Data analysis

The study employed two binomial logistic regression models (odds ratio = ORs, 95% CI) for the multivariate analysis. The first model was

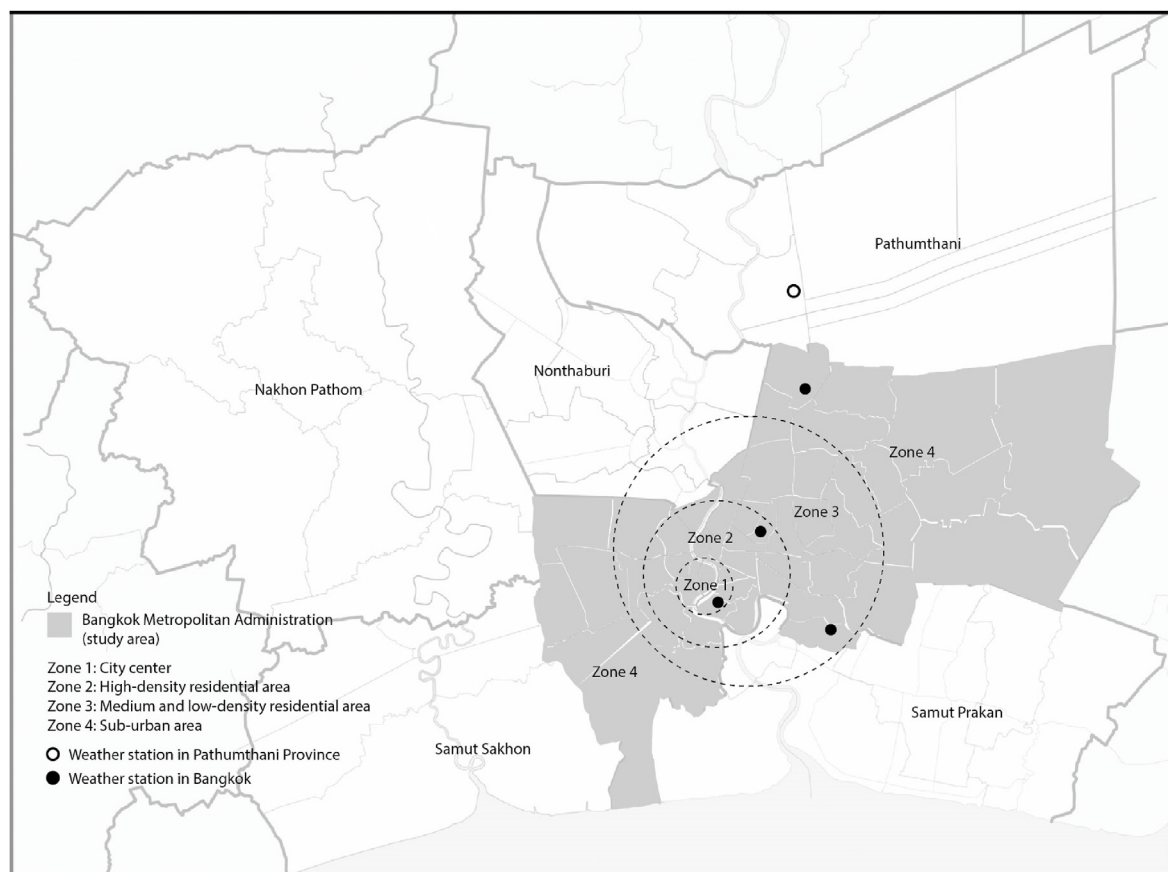


Fig. 2. Study area.

Table 1
Descriptive statistics of the respondents.

Category	Variable	Sample Characteristics (n = 505)
Socioeconomic Characteristics	Monthly income	
	Less than 5000 baht	8.7%
	5000–10,000	23.8%
	10,001–30,000	47.5%
	30,001–50,000	11.1%
	> 50,000 baht	8.9%
	Age (mean; SD)	39.69 (14.08)
	Education	
	Elementary	11.3%
	Secondary	30.7%
	Bachelor	51.5%
	Higher than Bachelor	6.5%
	Gender	
	Male	38.2%
	Female	61.8%
	Marital Status	
	Single	48.1%
	Married	47.9%
	Separated	4.0%
Urban environment	Air-conditioning at home	
	Yes	91.2%
	No	8.8%
	Air-conditioning at work	
	Yes	74.6%
	No	25.4%
	Park near home	
	< 1 km	58%
	1–3 km	16.4%
	3–5 km	9.3%
	> 5 km	16.2%
	UHI intensity	
	Low	26.9%
Health behavior	District population density	
	Low	24.4%
	Medium	28.5%
	High	47.1%
	Regular alcohol consumption	
	Yes	9.3%
	No	90.7%
	Regular smokers	
	Yes	6.3%
	No	93.7%
	Exercise 3 times a week	
	Yes	37.4%
	No	62.6%

to understand the association among the urban environment determinants, health behaviors, and socioeconomic conditions and self-reported heat stress, which was dichotomized into often and not often for analysis. The dependent variable for this model was four variables of heat stress: sleep, travel, work, and exercise. The second one was the health outcomes model, which established the association between exposure to heat stress and health outcomes. The dependent variable for this model was three variables of health outcomes: life satisfaction, energy level, and emotional problem.

3. Results

Characteristics of the respondents from the survey is presented in Table 1. We found that 47.3% of the respondents reported having a monthly income of 10,000–30,000 baht. The average age of the respondents was 39.6 years old, with mostly were single (48.1%), and

Table 2
Perceived heat stress in daily activities among respondents (in %).

	A little or none	Slightly	Moderate	Quite a lot	Almost every day
Urban Heat Stress Domains					
Sleeping	3.6	4.6	22.8	36.6	32.5
Daily travel	1.8	7.5	27.9	37.2	25.5
Work	2.8	7.1	31.5	36.2	22.4
Exercise	6.5	10.1	35.2	34.1	14.1

51.5% of the respondents had bachelor degrees. 61.8% of the respondents were female. More than half of the respondents (58%) reported living near the park or green open space. Regular cigarette or alcohol consumption was reported by 6.3% and 9.3%, respectively. Only 37.4% of the respondents reported being physically active. Most respondents reported having air-conditioning equipment at their home and office by 91.2% and 74.6%.

Table 2 presented the urban heat stress for four domains. Heat stress category ‘quite a lot’ was reported by 36.3% for sleeping, 37.2% for daily travel, 36.2% for work, and 34.1% for exercise. Less than 7% of respondents reported not having heat stress problems in their daily activities.

Table 3 presented the health and well-being outcomes. 39% of the respondents reported having moderate energy level problems, and 16.4% responded ‘quite a lot.’ Only 1% of respondents reported having an emotional problem ‘almost every day.’ Around 6% reported having ‘poor’ and ‘very poor’ life satisfaction.

Table 4 summarizes the association of heat stress domains with socioeconomic and urban environmental variables. The bivariate logistic regression, adjusting for potential confounders, shows several patterns and characteristics. Heat stress in four domains were increased with age (Sleeping OR = 0.910; daily travel OR = 0.944, work OR = 0.953, exercise OR = 0.955, $p < 0.05$). There is no consistent pattern of association between heat stress and income, marital status, and education. Healthy behaviors of the respondents were generally found not significant with heat stress. District population density and UHI intensity were found to be important confounders as they showed significant associations with all four domains of heat stress. Proximity

Table 3
Descriptive Statistics of Self-reported health outcomes.

Health and well-being outcomes	Proxy Question	%
Physical health (Energy level for daily activity)	How often did you have a problem with your energy level due to heat in the past for weeks?	
	a little or none	11.9%
	Slightly	32.1%
	Moderate	39.0%
	quite a lot	16.4%
Mental health (emotional problem)	almost every day	0.6%
	How often did you have a problem with your emotions due to heat in the past for weeks?	
	a little or none	13.3%
	Slightly	58.2%
	Moderate	31.1%
Well-being (life satisfaction)	quite a lot	5.5%
	almost every day	1.0%
	overall how would you rate your life satisfaction during the past four weeks?	
	Very poor	1.6%
	Poor	4.8%
	Fair	31.1%
	Good	49.5%
	Very good	13.1%

Table 4

The association of heat stress domains with socioeconomic and urban environmental variables.

Variable	OR (95% CI)			
	Sleeping	Daily Travel	Work	Exercise
Monthly income				
Less than 5000 baht	2.351 (1.762–2.621)*	5.333 (2.717–9.338)*	4.946 (4.002–6.129)*	1.423 (0.326–6.205)
5000–10,000	1.168 (1.317–2.762)*	2.646 (2.060–11.883)*	2.292 (1.039–2.6958)*	1.45 (0.410–5.119)
10,001–30,000	1.911 (0.437–8.346)	3.566 (1.931–13.653)*	2.117 (1.328–3.806)*	1.674 (0.535–5.231)
30,001–50,000	3.220 (0.355–13.886)	4.194 (2.964–6.8908)*	5.207 (0.833–32.523)	1.155 (0.304–4.386)
> 50,000 baht	ref	ref	ref	ref
Age	0.901 (0.834–0.950)*	0.944 (0.900–0.989)*	0.953 (0.915–0.993)*	0.955 (0.924–0.987)*
Education				
Elementary	0.669 (0.786–5.765)	3.450 (1.156–11.081)*	1.290 (0.01–1.982)	0.950 (0.461–1.438)
Secondary	0.775 (0.068–6.630)	3.670 (1.146–10.484)*	1.760 (0.178–12.194)	0.641 (0.319–1.848)
Bachelor	1.817 (0.258–7.756)	5.269 (0.205–23.053)	1.832 (0.265–4.44)	0.495 (0.503–1.739)
Higher than Bachelor	ref	ref	ref	ref
Gender				
Female	0.452 (0.177–0.553)*	1.903 (1.387–3.075)*	0.894 (0.378–2.111)	0.914 (0.457–1.826)
Male	ref	ref	ref	ref
Marital Status				
Single	2.271 (0.192–26.781)	0.860 (0.058–12.824)	0.560 (0.052–5.998)	0.422 (0.042–4.154)
Married	2.137 (0.202–22.514)	1.342 (0.095–18.870)	1.243 (0.121–12.75)	0.743 (0.078–7.062)
Separated	ref	ref	ref	ref
Air-conditioning at home				
No	2.154 (1.227–5.853)*	2.00 (0.298–13.339)	1.921 (1.401–9.869)*	2.892 (1.627–12.946)*
Yes	ref	ref	ref	ref
Air-conditioning at work				
No	0.298 (0.019–4.679)	1.925 (1.061–4.576)*	3.391 (2.026–5.688)	0.241 (0.03–1.909)
Yes	ref	ref	ref	ref
Park near home				
< 1 km	0.444 (0.239–0.797)*	3.003 (0.32–8.565)	0.849 (0.004–1.295)	0.363 (0.198–0.595)*
1–3 km	0.511 (0.206–0.832)*	3.225 (0.431–2.114)	1.040 (0.541–7.210)	0.812 (0.867–11.164)
3–5 km	0.594 (0.264–0.723)*	3.656 (1.942–3.925)*	1.908 (1.861–2.920)*	0.342 (0.673–8.145)
> 5 km	ref	ref	ref	ref
UHI intensity				
High	3.453 (2.675–7.438)*	3.213 (2.883–9.123)*	3.019 (1.998–5.893)*	3.792 (2.130–7.194)*
Medium	2.987 (1.987–6.978)*	2.194 (2.011–5.917)*	3.465 (2.579–6.721)*	2.106 (1.534–6.384)*
Low	ref	ref	ref	ref
District population Density				
High	1.431 (1.121–1981)*	1.876 (1.234–2.365)*	1.657 (1.395–1.875)*	1.276 (1.298–1.493)*
Medium	1.381 (1.190–1.682)*	1.619 (1.586–1.948)*	1.361 (1.270–1.946)*	1.171 (1.112–1.978)*
Low	ref	ref	ref	ref
Regular alcohol consumption				
No	0.263 (0.001–0.419)*	0.513 (0.132–1.991)	2.668 (0.537–13.238)	3.773 (0.823–17.298)
Yes	ref	ref	ref	ref
Regular smokers				
No	0.675 (0.041–1.519)	0.312 (0.132–1.991)	2.182 (0.534–11.283)	2.53 (1.233–3.298)*
Yes	ref	ref	ref	ref
Exercise 3 times a week				
No	0.62 (0.211–1.872)	0.654 (0.332–3.991)	3.365 (0.435–3.638)	2.473 (0.523–7.768)
Yes	ref	ref	ref	ref

Note: Model Sleeping: 2loglikelihood = 141.561, chi-square = 27.362, $p < 0.005$. Model daily travel: 2loglikelihood = 143.401, chi-square = 35.411, $p < 0.005$. Model work: 2loglikelihood = 479.310, chi-square = 29.289, $p < 0.05$. Model exercise: 2loglikelihood = 251.588, chi-square = 45.393, $p < 0.05$. * = $p < 0.05$.

to green open spaces also showed significantly associated with urban heat stress, although the effect was less pronounced. Elderly respondents were more likely to have heat stress problems in four domains of sleeping, daily travel, work, and exercise. Female respondents were more likely to experience heat stress during daily travel compared to men (OR = 1.903, $p < 0.05$).

Table 5 presented the association between heat stress and health outcomes. All three health outcomes were associated with four domains of health stress. Respondents who experience heat stress were more likely to have 'poor life satisfaction,' with the strongest association was found in 'exercise.' The association between physical health and heat stress domains was significant in 'travel,' 'work,' and 'exercise.' Well-being and mental health outcomes were significantly associated with all four domains of heat stress.

Note: Model Well-being: 2loglikelihood = 452.350, chi-square = 53.931, $p < 0.005$. Model Physical health: 2loglikelihood = 543.221, chi-square = 43.721, $p < 0.005$. Model Mental

health: 2loglikelihood = 657.101, chi-square = 36.975, $p < 0.005$. The model is adjusted with control variables: income, age, education, gender, marital status, air-conditioning at home, air-conditioning at work, park near home, UHI intensity, regular alcohol consumption, regular smokers, and exercise 3 times a week.

4. Discussion

The study represented the first report to understand the determinants of urban heat stress and its effects on human health in Bangkok, Thailand. Two main findings emerged from the study. First, urban heat stress was associated with socioeconomic conditions of the individual and urban environment determinants. People with less income were more likely to experience heat stress, sharing similarities with the previous study (Harlan et al., 2006). We found that age was associated with 'sleeping,' 'daily travel,' 'work' and 'exercise.' Age was widely considered as one of the major determinants of heat stress (Oudin

Table 5
Association between heat stress and health outcomes.

Variable	OR (95% CI)		
	Well-being	Physical health	Mental health
Sleeping			
A little or none	ref	ref	ref
Slightly	1.210 (0.153–0.414)	1.882 (0.598–5.923)	1.047 (0.314–2.853)
Moderate	1.374 (0.226–1.224)	1.649 (0.220–1.910)	1.227 (1.193–1.814)*
Quite a lot	1.426 (1.401–1.776)*	1.368 (0.240–1.738)	1.407 (1.296–1.870)*
Almost every day	1.655(0.982–2.790)*	1.647 (0.380–2.102)	1.517 (1.321–1.619)*
Travel			
A little or none	ref	ref	ref
Slightly	1.214 (0.149–9.879)	1.174 (0.116–4.994)	1.275 (0.114–3.977)
Moderate	1.582 (0.176–1.922)	1.452 (0.540–3.906)	1.663 (1.259–1.993)*
Quite a lot	1.497 (0.225–1.094)*	1.766 (1.378–2.553)*	1.716 (1.569–2.016)*
Almost every day	1.528 (0.263–1.062)*	1.885 (1.462–2.695)*	1.963 (1.538–1.723)*
Work			
A little or none	ref	ref	ref
Slightly	1.697 (0.933–2.641)	2.596 (0.977–11.683)	1.619 (0.114–3.977)
Moderate	1.422 (0.969–2.740)	1.143 (0.929–3.268)	1.993 (0.877–2.612)
Quite a lot	1.557 (0.691–2.512)	1.524 (0.916–3.242)	1.544 (1.275–1.717)*
Almost every day	1.783 (1.633–3.602)*	1.563 (1.417–1.756)*	1.790 (1.420–1.485)*
Exercise			
A little or none	ref	ref	ref
Slightly	1.360 (0.186–9.513)	1.72 (0.745–12.764)	1.489 (0.343–2.099)
Moderate	1.883 (1.401–10.763)*	1.291 (1.149–7.779)*	1.631 (0.278–1.434)
Quite a lot	1.929 (1.914–8.063)*	1.312 (1.909–4.536)*	1.759 (1.401–1.436)*
Almost every day	2.470 (1.294–7.713)*	1.327 (1.975–4.641)*	1.804 (1.754–2.613)*

Åström et al., 2011; Romero-Lankao et al., 2012). This is because when we get older, we have less heat tolerance (Robine et al., 2012). Although the effect was less pronounced, being female was more likely to experience heat stress during sleeping and daily travel, due to variations in body morphology. In other words, the female body had a smaller body and surface area that can adapt to heat loss compared to males (Notley et al., 2017). Low-income population groups were more likely to experience heat stress during daily travel. It was probably because they spent more time in traffic and outdoor during commute in hot periods due to traffic congestion. Although public transport options are available, the non-air conditioning public bus is still the most affordable alternative for low-income urban residents of Bangkok. Health behavior variables were found to have fewer effects on heat stress experienced in all domains. This was probably because only small percentages of the respondents reported consuming alcohol or cigarettes regularly (9.3% and 6.3% respectively).

We found that urban environment determinants were strong predictors for heat stress. UHI density was found to increase the likelihood of heat stress in all four domains. People living in high district population density were more likely to experience heat stress, as Zander et al. (2018b) have found in the Philippines. However, the effect was weaker in Bangkok. This is probably because of two things. First, the variations in the population density from the downtown to the suburban areas in Bangkok were too small. Second, this may also be because we employed the quota sampling for data collection.

Further investigations are needed to be able to establish the causation effect of population density on urban heat stress. Green open spaces were found to have alleviated the heat stress, especially at home. People living near parks and green spaces reported having less likelihood of experiencing heat stress during sleeping and exercise. This variable shows less pronounced association to heat stress, probably because Bangkok already has a low number green space area per person compared to WHO standard, around 5.5 sqm/person (Chandrasiri and Arifwidodo, 2017). It is interesting to notice that air-conditioning equipment was found less influential in relieving heat stress. We expected that people with air-conditioning at home would experience less heat stress in all domains. However, we found that it only helped to cope with sleeping, as previous studies have found (Okamoto-Mizuno

and Mizuno, 2012; Tokizawa et al., 2015). Having air-conditioning equipment at home was less beneficial for other heat stress domains. On the other hand, air-conditioning at work represented the type of jobs. We found that people with no air-conditioning at work were more likely to experience heat stress during work (OR = 3.391; 2.026–5.688), sharing similarities with previous studies (Zander et al., 2018a; Tawatsupa et al., 2014; Kjellstrom et al., 2018).

The second main finding of the study was that heat stress affects human health. In general, the association among heat stress domains and health outcomes echoed the previous studies (for example, see Guo et al., 2017; Wen Yi and Chan, 2015; Tawatsupa et al., 2012). We found that those who reported a higher level of heat stress to have adverse health and well-being outcomes (low life satisfaction, low energy level, and worse emotional problems). ORs of well-being in general (between 1.883 and 2.470) were worse than physical health (1.291 and 1.327) and mental health (1.759 and 1.804). Our result indicates that heat stress affected the well-being the most, which shares similarities with previous studies (Sekulova and van den Bergh, 2016; Osberghaus and Kühling, 2016). People who experienced heat stress at work were more likely to be associated with well-being, physical, and mental health problems. Previous studies have found that heat stress at work influences lower motivation to work, the overall well-being of workers, and contribute to lowering productivity and increased injury risk (Lan et al., 2010; Hancock et al., 2007). Heat stress during travel was also found associated with more likely to experience well-being, physical, and mental health problems. The psychological effects of heat stress have been found to impact mental health due to traffic congestion (Nadrian et al., 2019). The association of heat stress during exercise with all health outcomes in our result was similar to what has been established in the literature. Health and fitness were a major factor in heat stress risk at the individual level (Schuster et al., 2017). Under the hot environment, conducting physical activity and exercise can result in increased heat stress and exhaustion (Silva et al., 2019).

One limitation of the study is that we cannot be sure that individual and environmental determinants directly cause heat stress, and the health and well-being outcomes are caused directly by heat stress. Interpreting causality requires a more rigorous research method, such as using a longitudinal survey design with larger samples and better

probabilistic sampling. Another limitation is that the survey was conducted during the hot period (July–August). Different patterns may emerge when the survey is conducted during the cool season. Despite the limitations, the study contributes to the limited literature of heat stress and health outcomes at the urban scale using individual data. The survey design in our study enables the exploration of urban environmental determinants of heat stress as well as examines the effects of heat stress on urban residents. To our knowledge, this investigation is the first of its kind in Bangkok, Thailand. It is a first step towards bridging the gaps between research and practices in adaptation to global environmental changes at the urban scale.

The broader significance of the study is that it strengthens the argument that there is a need to address the adaptation strategy to heat stress at the city level. The standard strategy to adapt to heat stress at the individual level is to increase the number of air-conditioning units installed at home or workplace. As our study found out, it did not help with reducing heat stress during daily travel and exercise. This strategy can also be counter-productive as it will contribute further to increase UHI intensity as well as energy consumption (Arifwidodo, 2015). Instead, integrated strategies in urban planning and design policies are required. For example, protection and creation of green open spaces across the city will not only lower the effect of UHI intensity and air pollution but also reduce heat stress risks in the urban area (Laforteza et al., 2009). It is essential to integrate the green building initiatives into building codes and zoning policies, such as providing rooftop garden, floor area bonus for providing indoor green open spaces, and promoting the use of building materials that can reduce the temperature in the city (Sharma et al., 2018). It is also necessary to manage the heat risk through improved heat early warning systems and emergency response planning. Collaboration among stakeholders at the city and national levels is crucial to reduce the effects of heat stress in Bangkok.

5. Conclusion

The study explores two fundamental questions: (1) what are the determinants of urban heat stress in Bangkok? (2) are there any associations between urban heat stress and human health? These questions then used as the objectives of the study. By employing a comprehensive questionnaire, we could capture a detailed association of different domains of heat stress and their determinants. We have also been able to link heat stress and three measurements of health outcomes. We found that the socioeconomic conditions and urban environment were significant determinants of heat stress. However, some variables appeared to have different importance with what has been established in the literature. We also found that urban heat stress significantly affected human health. Most of the heat stress domains were significantly associated with health outcomes: well-being, physical health, and mental health. The findings suggest that the increased risk of heat stress represents a major problem in the urban area. As the temperature continues to increase due to climate change, the effects of heat stress are expected to worsen in the future. Future studies on urban heat stress are required, especially in cities of developing countries with tropical climate to understand different patterns and impacts to urban populations. Such research can inform local governments on how to formulate adaptation policies and measures.

CRediT authorship contribution statement

Sigit D. Arifwidodo: Conceptualization, Methodology, Formal analysis, Writing - review & editing. **Orana Chandrasiri:** Investigation, Project administration, Data curation, Writing - original draft.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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